Addressing the Confusion Between Hyperbolic Memory and Regime Switching in the Second Moment: A Markov Regime-Switching Hyperbolic GARCH Model[☆]

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Abstract

This paper studies the confusion between the Hyperbolic GARCH (HGARCH) and Markov Regime-Switching GARCH (MRS-GARCH) processes. We firstly propose the cause of this confusion and provide simulation evidence. An MRS-HGARCH model is then developed to address the confusion. Our Monte Carlo studies demonstrate that this model can effectively distinguish between the pure HGARCH and pure MRS-GARCH processes. When the proposed model is further employed to fit simulations containing both hyperbolic memory and regime switching, it outperforms the ordinary HGARCH model. Moreover, the related statistical properties including the stationarity conditions and asymptotic behaviours of the maximum likelihood estimators of the MRS-HGARCH process are briefly investigated. Finally, an empirical study of the S&P 500 index return is conducted which demonstrates that our MRS-HGARCH model can provide a more reliable estimator of the hyperbolic-memory parameter and outperform both the HGARCH and MRS-GARCH models.

JEL classification: C22; C51; C58; G32

Keywords: Stochastic Processes, Hyperbolic Memory, Regime Switching, Hyperbolic GARCH, MRS-HGARCH

1. Introduction

Long-memory persistence describes two important features of a scalar process: its spectral density at the origin is unbounded; and its autocorrelation function (ACF)

[☆]*Abbreviations:* GARCH, Generalized Autoregressive Conditional Heteroskedasticity; MRS, Markov Regime-Switching; ACF, autocorrelation function; ARFIMA, Autoregressive Fractionally Integrated Moving Average; DGP, data generation process; QMLE, Quasi Maximum Likelihood Estimation; TRTH, Thomson Reuters Tick History; BIC, Bayesian Information Criterion.

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decays at a hyperbolic rate at long lags (Qu, 2011). However, when regime switching is present (see Liu et al. (2011) and Foerster et al. (2016) for applications of this framework in economics research), a short-memory process can also demonstrate these features. This phenomenon is the so-called "spurious" long memory, which describes the confusion between long memory and regime switching (see, for example, Dieobold (1986), Perron (1989, 1990), Lamoureux and Lastrapes (1990), Lobato and Savin (1998), Diebold and Inoue (2001), Morana and Beltratti (2004), Perron and Qu (2010) and Qu (2014)). In particular, Diebold and Inoue (2001) propose a theoretical cause of this confusion for the first-moment case and argue that long memory and regime switching are interchangeable concepts and should not be studied separately.

Recent evidence suggests that the spurious long memory will be effectively addressed, if its cause can be appropriately controlled for. For instance, Perron and Qu (2010) and Qu (2011) develop powerful tests to distinguish between short- and long-memory processes in the first moment. Shi and Ho (2015) adopt a modelling approach to study this issue and propose the Markov Regime Switching Autoregressive Fractionally integrated Moving Average (MRS-ARFIMA) model. Their simulation results demonstrate that such a model can effectively distinguish between the pure MRS and ARFIMA data generate processes (DGPs). Additionally, a consistent estimator of the long-memory parameter (d) can be simultaneously provided, which complements the testing approach adopted by Perron and Qu (2010) and Qu (2011).

This paper aims to extend the research of Shi and Ho (2015) to investigate the second-moment cases. Among the existing second-moment models, GARCH-type models (Bollerslev, 1986) have enjoyed great popularity because of their ability to capture typical properties such as time-varying heteroskedasticity and volatility clustering (French et al., 1987; Franses and van Dijk, 1996; Bauwens et al., 2006; Ho et al., 2013; Canova and Pérez Forero, 2015). To study the long-memory persistence within the GARCH framework, the Fractionally Integrated GARCH (FIGARCH) model proposed by Baillie et al. (1996) is widely employed. Analogous to ARFIMA, this model employs the fractional differencing operator to study the autoregressive structure of the conditional variance (Baillie et al., 1996).

However, as pointed by Giraitis et al. (2000), a general covariance stationary ARCH(∞) process always has absolutely summable autocovariances. Davidson (2004) further suggests that the true long memory in variance cannot be represented by the ARCH(∞) framework. Hence, to measure the memory of an ARCH(∞) process

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more precisely, Davidson (2004) uses the concept of hyperbolic memory. A process is said to have hyperbolic (geometric) memory if its ACF decays hyperbolically (geometrically). Thus, related GARCH-type model like FIGARCH actually describes the hyperbolic-memory persistence of the conditional variance. d is therefore referred to as the hyperbolic-memory parameter.

The main problem of the FIGARCH model is that it does not have a finite unconditional variance (Davidson, 2004; Conrad, 2006; Li et al., 2015). To overcome this issue, we adopt the recently proposed Hyperbolic GARCH (HGARCH) model (Li et al., 2015), which nests the GARCH and FIGARCH as two special cases. Compared with a similar model (HYGARCH) developed by Davidson (2004), HGARCH is a simplified but powerful version. In terms of studying the regime switching in the second moment, various Markov Regime-Switching GARCH (MRS-GARCH) models are constructed from the seminal work of Hamilton (1989). We employ the one discussed in Haas et al. (2004) because that it outperforms other specifications by appropriately modelling the path dependency of the conditional variance.

Spurious hyperbolic memory in the second moment is found in various studies (see, for example, Mikosch and Starica (2003) and Lai and Xing (2006)). For the MRS-GARCH process, we propose that this is due to the dependency of transition probabilities on the sample size. This result complies with that of Diebold and Inoue (2001). To verify our proposition, we perform a Monte Carlo study fitting the simulated MRS-GARCH data by the HGARCH model. Consistent with the existing evidence, our simulation results suggest that the MRS-GARCH DGP can be significantly confused with a hyperbolic-memory process like HGARCH.

Motivated by Shi and Ho (2015), we then aim to address this confusion by extending their model to the second moment. An MRS-HGARCH framework is therefore proposed with brief discussions on its statistical properties such as stationarity conditions and existence of higher moments. To demonstrate its effectiveness, we perform another Monte Carlo study with three DGPs: pure HGARCH, pure MRS-GARCH and MRS-HGARCH. When fitted by our proposed model, simulations of the pure MRS-GARCH DGP (pure HGARCH and MRS-HGARCH DGPs) lead to insignificant (significant) hyperbolic memory. Also, the mean estimates of d are close to their true values for the pure HGARCH and MRS-HGARCH DGPs. Therefore, the proposed MRS-HGARCH model can effectively distinguish between hyperbolic memory and regime switching and provide a consistent estimator of d in the second moment.

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To empirically compare the model performance, we fit the HGARCH, MRS-GARCH and MRS-HGARCH frameworks for the daily returns of the S&P 500 index ranging from 1 January 2001 to 31 December 2015. It is demonstrated that the identified structures of the underlying volatility states of the MRS-HGARCH are similar to those of the MRS-GARCH model. Their patterns are consistent with the real trend of the macro-economy. Compared with the HGARCH model, MRS-HGARCH can generate potentially more reliable estimates of d. In terms of model evaluations and forecasting performance, MRS-HGARCH beats both the HGARCH and MRS-GARCH models. Nevertheless, all of them show acceptable residual diagnostics results.

The contributions of this paper to the existing literature are discussed below. First, related research like Beine et al. (2001), Lux and Morales-Arias (2010) and Raggi and Bordignon (2012) has allowed for the regime-switching parameters within the hyperbolic-memory GARCH models. Although most of those studies mention that the estimated d via this approach is potentially more reliable, no evidence or analysis is provided. In this study, we firstly propose a theoretical cause of the spurious hyperbolic memory in the second moment. Our simulation evidence further demonstrates that this cause can be properly controlled by allowing for the regime-switching parameters. Hence, our results significantly complement the relevant studies and support the reliability of estimates obtained from such a parametric approach. Second, evidenced by simulation results, our proposed MRS-HGARCH model can effectively distinguish between the pure MRS-GARCH and pure HGARCH DGPs. Although related powerful tests have been proposed by Perron and Qu (2010) and Qu (2011), they focus on the first moment, and it is unknown if those tests can be extended to the second-moment case directly. Additionally, our approach can provide a consistent estimator of d under various circumstances, which is not achievable using tests. Thus, our study sheds some light on addressing the spurious hyperbolic memory in the second moment. Next, the statistical properties like existence of unique solution, stationarity and higher moments of the MRS-HGARCH process are investigated and presented. The asymptotic properties of the corresponding maximum likelihood estimators are also briefly discussed. Finally, our empirical study shows that significant hyperbolic memory still exists when the effects of regime switching are already considered. This result demonstrates the importance of considering the hyperbolic memory when studying the financial volatility in practice.

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2. Hyperbolic memory and the related GARCH models

As described in Diebold and Inoue (2001), long memory is defined using the rate of growth variances of partial sums as $var(S_T) = O(T^{2d+1})$, where $S_T = \sum_{t=1}^{T} r_t$, $\{r_t\}$ is a sequence of interested financial series, and T is the number of observations. Since it cannot exist in a covariance stationary ARCH(∞) process (Giraitis et al., 2000), Davidson (2004) suggests using the concept of hyperbolic memory instead. Defining that $h_t = \omega + \sum_{i=1}^{\infty} \theta_i$, an ARCH($\infty$) process r_t has hyperbolic memory in its second moment if $\theta_i = O(i^{-1-d})$, where ω is a non-negative constant, and h_t is the conditional variance of r_t. In particular, d is referred to as the hyperbolic-memory parameter.

For the study of time-varying financial volatility, the FIGARCH model is widely employed to investigate its long-range dependence. Extended from the original GARCH model, the FIGARCH(p,d,q) model is proposed by Baillie et al. (1996) with the following specification:

$$\begin{aligned} r_t &= \mu + \varepsilon_t \\ \varepsilon_t &= \eta_t \sqrt{h_t} \text{ where } \eta_t \overset{iid}{\sim} N(0,1) \\ h_t &= \frac{\omega}{\beta(1)} + \left[1 - \frac{\phi(L)}{\beta(L)}(1-L)^d\right] \varepsilon_t^2 \end{aligned} \tag{1}$$

where ε_t is the residual at time t, η_t is an identical and independent sequence following a Gaussian distribution, L is the lag operator, $\phi(L) = 1 - \sum_{i=1}^{q} \phi_i L^j$, $\beta(L) = 1 - \sum_{j=1}^{p} \beta_j L^i$, and $(1-L)^d$ is the fractional differencing operator as defined by Hosking (1981):

$$(1-L)^d = \sum_{k=0}^{\infty} \delta_k(d) L^k,\ \delta_k(d) = \frac{k-1-d}{k}\delta_{k-1}(d) \text{ and } \delta_0(d) = 1 \tag{2}$$

In particular, at time t, we follow Baillie et al. (1996), Bollerslev and Mikkelsen (1996) and Li et al. (2015) and proxy this infinity process by $(1-L)^d = \sum_{k=0}^{1000} \delta_k(d) L^k$. We have a stationary FIGARCH process when $0 < d < 1$. If $d = 1$, the process has a unit root and thus a permanent shock effect, which is equivalent to the IGARCH model (Baillie et al., 1996). If $d = 0$, the process reduces to an ordinary GARCH process.

Despite its attractive features, FIGARCH model has many well-known drawbacks, such as the infinite unconditional variance (Davidson, 2004; Conrad, 2006; Li et al., 2015). In order to overcome those problems, Davidson (2004) proposes a hyperbolic GARCH (HYGARCH) model with the following conditional variance equation:

$$h_t = \frac{\omega}{\beta(1)} + \left\{1 - \frac{\phi(L)}{\beta(L)}\left[1 - \gamma + \gamma(1-L)^d\right]\right\} \varepsilon_t^2 \tag{3}$$

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As discussed in Li et al. (2015), the HYGARCH model can be rewritten as a weighted average of the GARCH and FIGARCH process. It reduces to the GARCH (FIGARCH) when $\gamma = 0$ ($\gamma = 1$). The HYGARCH process now has a finite unconditional variance when $1 - (1 - \gamma)\phi(1)/\beta(1) < 1$. Despite this, its conditional variance may have an unnecessarily complicated form, and both parameters d and γ are related to its memory measure (Li et al., 2015). Motivated by the HYGARCH specification, Li et al. (2015) recently propose a new hyperbolic GARCH (HGARCH) with the following conditional variance equation:

$$h_t = \frac{\omega}{\beta(1)} + \gamma \left[1 - \frac{\phi(L)}{\beta(L)}(1 - L)^d\right] \varepsilon_t^2 \tag{4}$$

where $0 < d \leq 1$, $\omega > 0$ and $\gamma > 0$. When $d = 1$, from its evolution, the HGARCH model becomes a general GARCH model with geometric memory and an IGARCH model if it is further restricted that $\gamma = 1$. If we only set $\gamma = 1$, the HGARCH model will reduce to a FIGARCH model. The HGARCH process has a hyperbolic memory if $0 < d < 1$, and the hyperbolic decays are more significant when d is closer to 0. For the non-negativity conditions of the HGARCH model, they are independent of γ. The restrictions on d, ϕ_js and β_is, are exactly the same as those of the FIGARCH model discussed in Conrad and Haag (2006). Hence, they are less complicated than those of the HYGARCH model, which are examined in Conrad (2010). Further, Equation (4) can be rewritten into the form of ARCH(∞) as

$$h_t = \frac{\omega}{\beta(1)} + \sum_{k=1}^{\infty} b_k \varepsilon_{t-k}^2 \tag{5}$$

where b_ks depend on the values of ϕ_js, β_is and d. Also, Li et al. (2015) argue that it can be verified that $\gamma = \sum_{k=1}^{\infty} b_k$. Therefore, two important implications can be derived: the unconditional variance of the HGARCH process is $\omega/[(1 - \beta_1)(1 - \gamma)]$, which is finite given that $0 < \gamma < 1$;[1] and the decaying pattern of b_ks is controlled by d. In other words, γ is the amplitude parameter, while d measures the memory. Because of its comparatively simpler structure, we will employ the HGARCH model to study the hyperbolic memory for the second-moment cases in the rest of this paper. Regarding the estimation, similar to the GARCH model, the quasi-maximum likelihood estimation (QMLE) can be used. When relevant assumptions in Li et al. (2015) are satisfied, such estimators are asymptotically consistent and normally distributed. This is because that HGARCH model falls in the fractional GARCH case studied by Robinson et al. (2006),

[1] To ensure the non-negativity of h_t, it is already constrained that $\omega > 0$ and $\beta_1 < 1$, see Conrad and Haag (2006) for details.

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for which the asymptotic properties are well established.

We notice that in Equation (1), the distribution of innovation is assumed to be Gaussian. However, significant evidence suggests that the financial series is rarely Gaussian but typically leptokurtic and exhibits heavy-tail behaviour (Bollerslev, 1987; Susmel and Engle, 1994). In addition, Student's t-distribution is a widely used alternative, which can accommodate the excess kurtosis of the innovations (Bollerslev, 1987). Moreover, despite its the consistency, the QML estimators with the Gaussian distribution for GARCH family models is generally less efficient than those of the ML with Student's t-distribution in practice (Susmel and Engle, 1994). Therefore, the Student's t-distribution will be adopted for the HGARCH model in the rest of this study.

3. MRS-GARCH model

The main weakness of the GARCH family model is that it assumes that the conditional variance has only one regime over the entire period. To control for the possible structural changes, Hamilton (1989) proposes the MRS model, which allows the jumps between state spaces. To retain the advantages of GARCH models and allow the structure breaks at the same time, Cai (1994) and Hamilton and Susmel (1994) apply this seminal idea to the ARCH specification. However, the extension to the GARCH specification is difficult because of the path dependency of the conditional variance term. Researchers such as Gray (1996), Dueker (1997), Lin (1998) and Klaassen (2002) have generated various algorithms to overcome this issue.

In this paper, we employ the N-state MRS-GARCH(1,1) model with the Student's t-innovations investigated in Haas et al. (2004) as follows:[2]

$$\begin{aligned} r_t &= \mu + \varepsilon_t \\ \varepsilon_t &= \eta_t \sqrt{h_t} \text{ where } \eta_t \overset{iid}{\sim} t(0,1,v) \\ h_{s_t,t} &= \omega_{s_t} + \alpha_{s_t}\varepsilon_{t-1}^2 + \beta_{s_t} h_{s_t,t-1} \end{aligned} \tag{6}$$

[2]The reason for using this specification rather than those studied in Gray (1996), Klaassen (2002) and others is the path dependency problem of the MRS-GARCH-type models. In Haas et al. (2004), this is solved by assuming N independent GARCH sequences, and they will only be recombined in the construction of likelihood. In contrast, Gray (1996), Klaassen (2002) and other similar studies try to employ certain recombining methods to generate an overall GARCH sequence. It may cause difficulty in interpreting the meaning of system parameters, since current conditional variance will be affected by past values from all states. Details can be found in Haas et al. (2004), and the authors refer to this as the analytical intractability problem. To avoid such issues, we adopt the MRS-GARCH specification proposed by Haas et al. (2004) in this study.

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$$h_t = \sum_{n=1}^{N} I(s_t = n) h_{n,t}$$

where s_t is the underlying volatility state that the r_t lies in at time t, η_t is an identical and independent sequence following a Student's t-distribution with mean of 0 and standard deviation of 1, v is the degree of freedom of the Student's t-distribution, $h_{s_t,t}$ is the conditional variance in state s_t at time t, and $I(\cdot)$ is the indication function. Further, the sequence $\{s_t\}$ is assumed to be a stationary, irreducible Markov process with discrete state space $\{1, 2, \ldots, N\}$ and transition matrix $P = [p_{jk}]$ where $p_{jk} = P(s_{t+1} = k | s_t = j)$ is the transition probability of moving from state j to state k ($j, k \in \{1, 2, \ldots, N\}$).

To the best of our knowledge, there is no effective test developed so far to determine the optimal number of states N. Studies like Cho and White (2007) can only test the null hypothesis of one regime against the alternative of regime switching between two states. A recent research of Faias and Nunes (2012) comprehensively reviews the existing regime-switching tests, but none of them test the case of three or more states. Other literature like Carrasco et al. (2014) tests parameters of the MRS model rather than the number of states, which is irrelevant to this purpose. Furthermore, the classic trinity (Wald, likelihood ratio and score) tests are not applicable here, due to the fact that the MRS-type model with more states does not nest the one with fewer states (Wilfling, 2009). On the other hand, although there are many existing tests examining the number and/or locations of the structural changes (such as the non-parametric test used in Ross (2013)), it is not straightforward to extend them to the MRS cases. Another potential problem is that even if the optimal number of states can be tested, their economic interpretation might not be straightforward. Hamilton's (1989) original work suggests that the two-state case can be linked to the concept of business cycle. For instance, low- and high-mean states can be referred to as the trough and peak periods, respectively. However, the interpretation for a large number of such states might not be meaningful. Due to those reasons, we only consider the two-state case for all the MRS-type and related models in this paper.

Under fairly general conditions, Haas et al. (2004) and Haas and Paolella (2012) demonstrate the existence of the overall ($E(\varepsilon_t^2) = E(h_t)$) and state-specific ($E(\varepsilon_t^2 | s_t = n) = E(h_{n,t})$) unconditional variance of an MRS-GARCH process. To differentiate the two states considered in this paper, we define states 1 and 2 as calm (low-volatility) and turbulent (high-volatility), respectively. Hence, it is constrained that $E(h_{1,t}) < E(h_{2,t})$, and the formulas to calculate those values can be found in Haas et al. (2004) and Haas

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and Paolella (2012).

We estimate the parameters of the two-state MRS-GARCH model using MLE. The conditional density of ε_t is given by Mullen et al. (2011) as:

$$\begin{aligned} \Omega_{t-1} &= \{\varepsilon_{s_t,t-1}, \varepsilon_{s_t,t-2}, ..., \varepsilon_{s_t,1},\} \\ \theta &= (\mu, \omega_1, \omega_2, \alpha_1, \alpha_2, \beta_1, \beta_2, p_{11}, p_{22}, v)' \\ f(\varepsilon_{s_t,t}|s_t &= j, \theta, \Omega_{t-1}) = \frac{\Gamma[(v+1)/2]}{\Gamma(v/2)\sqrt{\pi(v-2)h_{j,t}}} \left[1 + \frac{\varepsilon_{j,t}^2}{(v_j-2)h_{j,t}}\right]^{\frac{v+1}{2}} \end{aligned} \tag{7}$$

Plugging the filtered probability in state j at time $t-1$, $\rho_{j,t-1} = P(s_{t-1} = j|\theta, \Omega_{t-1})$, into Equation (7) and integrating out the state variable s_{t-1}, the density function in Equation (7) becomes:

$$f(\varepsilon_{s_t,t}|\theta, \Omega_{t-1}) = \sum_{j=1}^{2}\sum_{k=1}^{2} p_{jk}\rho_{j,t-1} f(\varepsilon_{s_t,t}|s_t = j, \theta, \Omega_{t-1}). \tag{8}$$

$\rho_{j,t-1}$ can be obtained using an integrative algorithm given in Hamilton (1989). The log-likelihood function corresponds to Equation (6) as follows:

$$L(\theta|\varepsilon) = \sum_{t=2}^{T} \ln f(\varepsilon_{s_t,t}|\theta, \Omega_{t-1}), \text{ where } \varepsilon = (\varepsilon_{s_t,1}, \varepsilon_{s_t,2}, ..., \varepsilon_{s_t,T})', \tag{9}$$

and the ML estimator $\hat{\theta}$ is obtained by maximizing Equation (9).

To identify the underlying volatility state sequence $\{s_t\}$, the smoothing probability of the calm state is constructed as follows (Hamilton, 1989):

$$P(s_t = 1|\theta, \Omega_T) = \rho_{1,t}[\frac{p_{11}P(s_{t+1} = 1|\theta, \Omega_T)}{P(s_{t+1} = 1|\theta, \Omega_t)} + \frac{p_{12}P(s_{t+1} = 2|\theta, \Omega_T)}{P(s_{t+1} = 2|\theta, \Omega_t)}] \tag{10}$$

Using the fact that $P(s_T = 1|\theta, \Omega_T) = \rho_{1,T}$, the smoothing probability series $P(s_t = 1|\theta, \Omega_T)$ can be generated by iterating Equation (10) backwards from T to 1. As suggested by Hamilton (1989), a widely recognized rule to identify the state of r_t is that if $P(s_t = 1|\theta, \Omega_T)$ is less than 0.5, r_t is assumed to lie in the turbulent state at time t and otherwise in the calm state.

In addition to the discussion at the end of Section 3, we do not use QMLE with Gaussian distribution in Equation (6) for the following reasons. As noted by Klaassen (2002), Ardia (2009) and Haas (2009), if regimes are not Gaussian but leptokurtic, the use of within-regime normality can seriously affect the identification of the regime process. The details can be found in Haas and Paolella (2012), who further argue that

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the QMLE based on Gaussian components does not provide a consistent estimator of the MRS-GARCH model if the true distribution of innovations is not Gaussian. Moreover, a recent study by Shi and Feng (2016) provide simulation evidence to support this argument. Hence, we use the assumption of the Student's t-distribution for all the MRS-type models in the rest of this paper.

4. Confusion between hyperbolic memory and regime switching

Diebold and Inoue (2001) theoretically analyse the confusion between long memory and regime switching in the first moment and argue that it is due to the dependence of p_{11} and p_{22} on T. We hereby propose that this dependence is also the cause of the hyperbolic-memory behaviour for the MRS-GARCH process.[3]

Proposition 1. *For the following two-state MRS-GARCH(1,1) model:*

$$
\begin{aligned}
r_t &= \mu + e_t \\
e_t &= I(s_t = 1)\varepsilon_{1,t} + I(s_t = 2)\varepsilon_{2,t} \\
\varepsilon_{1,t} &= \eta_{1,t}\sqrt{(h_{1,t})} \text{ and } \varepsilon_{2,t} = \eta_{2,t}\sqrt{(h_{2,t})}, \text{ where } \eta_{1,t}, \eta_{2,t} \overset{iid}{\sim} N(0,1) \\
h_{1,t} &= \omega_1 + \alpha_1 \varepsilon_{1,t-1}^2 + \beta_1 h_{1,t-1} \\
h_{2,t} &= \omega_2 + \alpha_2 \varepsilon_{2,t-1}^2 + \beta_2 h_{2,t-1} \\
P &= \begin{pmatrix} p_{11} & 1-p_{11} \\ 1-p_{22} & p_{22} \end{pmatrix}
\end{aligned}
$$

where $I(\cdot)$ is the indication function, P is the transition probability matrix and $\{s_t\}$ is an irreducible and stationary Markov process with state space {1,2}. Assume that $\{s_t\}$, $\{\eta_{1,t}\}$, $\{\eta_{2,t}\}$, $\{h_{1,t}\}$ and $\{h_{2,t}\}$ are independent for all $t \leq T$, $p_{11} = 1 - c_1 T^{-\delta_1}$ and $p_{22} = 1 - c_2 T^{-\delta_2}$, with δ_1, $\delta_2 > 1$ and $0 < c_1$, $c_2 < 1$. Further, let α_1, α_2, β_1, $\beta_2 > 0$, $\alpha_1 + \beta_1 < 1$, $\alpha_2 + \beta_2 < 1$, $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$ and $3\alpha_2^2 + 2\alpha_2\beta_2 + \beta_2^2 < 1$, that is, $\varepsilon_{1,t}$, $\varepsilon_{2,t}$, $\varepsilon_{1,t}^2$ and $\varepsilon_{2,t}^2$ are covariance stationary. Then the variance of partial sums of e_t^2 grows at a rate corresponding to $I((1/2)max(0, min(\delta_1, \delta_2) - |\delta_1 - \delta_2| - 1)$.

Proof. See Appendix A.

Hence, it is expected that a pure MRS-GARCH process can have hyperbolic decays

[3]The specification considered in the Proposition 1 is only employed for demonstration purpose. More complex situations such as the non-normally distributed disturbance, leverage in GARCH, ARMA structure in the mean equation and multi-state regimes can be explored using similar ideas to that presented in the proof.

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in the ACF of its squared residuals[4], which may be confused with a pure HGARCH process. We demonstrate this confusion with simulation evidence using HGARCH(1,d,1) and MRS-GARCH(1,1) DGPs.[5] Altogether, nine sets of simulations with different d, ϕ and β are generated for the HGARCH(1,d,1) process, where $\mu = 0$, $\omega = 0.1$, $\gamma = 0.5$ and $T = 5,000$. For the two-state MRS-GARCH(1,1) process, we construct twelve sets of different p_{11}, p_{22}, P_1 and P_2, where $\mu = 0$, $\omega_1 = 0.1$, $\omega_2 = 0.5$ and $T = 5,000$.[6] In addition, when volatility persistence P is 0.7, α (β) is set to 0.2 (0.5); when P is 0.9, α (β) is set to 0.3 (0.6); and when P is 0.8, α (β) is set to 0.25 (0.55). Moreover, simulated data of the HGARCH and MRS-GARCH processes follow a Student's t-distribution with 3 degrees of freedom. To avoid the starting bias, 10,000 points are generated for each simulation, and then the first 5,000 are truncated. Finally, 300 replicates are produced for each set of parameters. We fit the HGARCH(1,d,1) models for both the generated HGARCH and MRS-GARCH simulations, and the results are presented in Table 1.

In Table 1, it can be seen that when the true DGP is HGARCH, the mean estimates of d are close to the true values with small standard errors (SEs). However, when the true DGP is MRS-GARCH, the mean estimates of d obtained from the HGARCH model are also considerably small. Additionally, the fractions of rejection of the geometric memory ($H_0 : d = 1$ against $H_0 : d < 1$, see Davidson (2004)) at 5% are greater than 0.95 in all cases of HGARCH and MRS-GARCH DGPs.[7] Hence, it suggests that MRS-GARCH DGP can be significantly confused with HGARCH DGP.

[4]Our proposition suggests that a covariance stationary MRS-GARCH process can have "true" long memory under similar assumptions used by Diebold and Inoue (2001). This does not contradict the findings in Giraitis et al. (2000), as the MRS-GARCH cannot be nested by the ARCH(∞) family covered in their research. In other words, although MRS-GARCH process is not designed to study the long memory, it could have a "true" long memory. Therefore, this process will have hyperbolic decays in the ACF of its squared residuals, although this is not expected.

[5]As evidenced by many economic and finance studies (see, for example, Hansen and Lunde (2005)) the more sophisticated specifications do not outperform the simple GARCH(1,1) specification for criteria like the out-of-sample forecast. Hence, we only consider the (1,1) case for all GARCH-type models employed in this paper.

[6]In Diebold and Inoue (2001), the confusion between long memory and regime switching is much more significant when transition probabilities are greater than or equal to 0.99. Thus, we only select the transition probabilities studied in their work that are greater than or equal to 0.99. For the same reason, we set T to be 5,000 for both HGARCH and MRS-GARCH processes. We have also investigate the case where transition probabilities are smaller (e.g. 0.90 and 0.95), and the results are generally consistent.

[7]To be consistent with the results in Section 5, we perform the likelihood-ratio test (LRT) to examine the significance of the hyperbolic memory by comparing the likelihoods of the fitted GARCH and HGARCH models. Since the ML estimator of HGARCH has well-defined asymptotic properties, the LRT will be asymptotically equivalent to the score test proposed by Li et al. (2015).

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Table 1
Estimates of HGARCH(1,d,1) Models

(a) Simulated MRS-GARCH(1,1) Data

p_{11}	p_{22}	P_1	P_2	$Mean_d$	SE_d	$F.R.$
0.99	0.999	0.7	0.9	0.7825	0.1726	0.9633
		0.8	0.8	0.7332	0.2215	0.9767
		0.9	0.7	0.8081	0.1876	0.9633
0.999	0.99	0.7	0.9	0.7629	0.1433	0.9633
		0.8	0.8	0.8086	0.1386	0.9567
		0.9	0.7	0.8967	0.0975	0.9533
0.99	0.99	0.7	0.9	0.8688	0.0990	0.9533
		0.8	0.8	0.7577	0.1347	0.9833
		0.9	0.7	0.7745	0.1554	0.9767
0.999	0.999	0.7	0.9	0.5961	0.1477	0.9867
		0.8	0.8	0.5135	0.1922	0.9933
		0.9	0.7	0.6102	0.2497	0.9867

(b) Simulated HGARCH(1,d,1) Data

d	ϕ	β	$Mean_d$	SE_d	$F.R.$
0.45	0.10	0.20	0.4574	0.1284	1.0000
	0.15	0.15	0.4787	0.1343	1.0000
	0.20	0.10	0.4624	0.1358	1.0000
0.55	0.10	0.20	0.5579	0.1300	1.0000
	0.15	0.15	0.5782	0.1347	1.0000
	0.20	0.10	0.5661	0.1302	1.0000
0.60	0.10	0.20	0.6126	0.1253	1.0000
	0.15	0.15	0.6006	0.1323	1.0000
	0.20	0.10	0.6164	0.1303	1.0000

Note: P_1 is the volatility persistence of calm state ($P_1 = \alpha_1 + \beta_1$). P_2 is the volatility persistence of turbulent state ($P_2 = \alpha_2 + \beta_2$). The Monte Carlo study is based on 300 replications, and the sample size is 5,000. $Mean$ is the mean of estimates. SE is their standard error. $F.R.$ is the fraction of rejection with the hypotheses $H_0 : d = 1$ against $H_1 : d < 1$ at 5% level.

5. MRS-HGARCH model

Shi and Ho (2015) propose an MRS-ARFIMA model and demonstrate its effectiveness to control for the impacts of the transition probabilities, which is the cause of the spurious long memory in the first moment (Diebold and Inoue, 2001). As shown in Section 4, such impacts can also lead to the spurious hyperbolic memory for an MRS-GARCH process. Therefore, it is expected that adopting the similar modelling structure proposed by Shi and Ho (2015) can also effectively distinguish between the pure HGARCH and pure MRS-GARCH DGPs. Based on this idea, we follow Haas et al.

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(2004) to construct the regime-switching parametric structure within the HGARCH framework. The MRS-HGARCH model with Student's t-distribution is then proposed as follows.[8]

1

$$\begin{aligned} r_t &= \mu + \varepsilon_t \\ \varepsilon_t &= \eta_t \sqrt{h_t} \text{ where } \eta_t \overset{iid}{\sim} t(0,1,v) \\ h_{s_t,t} &= \frac{\omega_{s_t}}{\beta_{s_t}(1)} + \gamma_{s_t}\left[1 - \frac{\phi_{s_t}(L)}{\beta_{s_t}(L)}(1-L)^{d_{s_t}}\right]\varepsilon_t^2 \\ h_t &= \sum_{n=1}^{N} I(s_t = n) h_{n,t} \end{aligned} \tag{11}$$

where ω, ϕ_is, β_js, γ and d can switch between states, and s_t is defined in the same way as in Equation (6). Following Equation (5), its conditional variance equation can be rewritten as

$$h_t = \sum_{n=1}^{N}[I(s_t = n)\frac{\omega_n}{\beta_n(1)}] + \sum_{k=1}^{\infty}\left\{\sum_{n=1}^{N}[I(s_t = n)b_{n,k}]\varepsilon_{t-k}^2\right\} \tag{12}$$

where $b_{n,k}$ depends on the values of $\phi_{n,i}$s, $\beta_{n,j}$s and d_ns.

Further, we notice that the coefficients in Equation (12) depend on time t and it can be rewritten as follows.

$$\begin{aligned} &h_t = a_0(t) + \sum_{k=1}^{\infty} a_k(t)\varepsilon_{t-k}^2, \text{ where} \\ &a_0(t) = \sum_{n=1}^{N}[I(s_t = n)\frac{\omega_n}{\beta_n(1)}],\ a_k(t) = \sum_{n=1}^{N}[I(s_t = n)b_{n,k}] \text{ and } \sum_{n=1}^{N}[I(s_t = n)] = 1. \end{aligned} \tag{13}$$

Hence, the MRS-HGARCH belongs to the nonrescaled time-varying ARCH(∞) process studied in Dahlhaus et al. (2006). Using their well-established results, the following facts can be derived.

Theorem 1. *Suppose that $\phi_{n,i}s$, $\beta_{n,j}s$ and d_n satisfy the non-negativity constraints in Conrad and Haag (2006), and $\sum_{k=1}^{\infty}\left[\sup_n b_{n,k}\right] < 1$. Then there exists an almost surely well-defined unique solution in the set of all causal solutions of the MRS-HGARCH model defined by Equation (11) with $E(\varepsilon_t^2) < \infty$. The conditional variance has the form*

[8] Following the original work of Haas et al. (2004), the μ and v in Equation (11) are not allowed to be regime-switching. Examination of a more flexible MRS-HGARCH process allowing such features is beyond the scope of this paper and remains for future works.

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of a time-varying Volterra series expansion as follows

$$h_t = a_0(t) + \sum_{k=1}^{\infty} \sum_{j_1,\ldots,j_k \geq 1} \left[a_0\Big(t - \sum_{s=1}^{k} j_s\Big) \right] \prod_{r=1}^{k} \left[a_{j_r}\Big(t - \sum_{s=1}^{r-1} j_s\Big) \right] \prod_{r=1}^{k} \eta^2_{t-\sum_{s=1}^{r} j_s} \tag{14}$$

where $a_k(t)$ is defined in Equation (13) for all $k \geq 0$. If we further have that $p_{ij} \in (0,1)$ for all $i, j \in \{1, 2, \ldots, N\}$, $E(\varepsilon_t^2)$ will not depend on t.

Proof. See Appendix B.

As for the stationarity conditions of the MRS-HGARCH process, we have the following fact.

Theorem 2. *Suppose that $\phi_{n,i}$s, $\beta_{n,j}$s and d_n satisfy the non-negativity constraints in Conrad and Haag (2006), $p_{ij} \in (0,1)$ for all $i, j \in \{1, 2, \ldots, N\}$, $E(\eta_t^4) < \infty$ and $\sum_{k=1}^{\infty} \left[\sup_n b_{n,k} \right] < \left[E(\eta_t^4) \right]^{-1/2}$, then $E(\varepsilon_t^4) < \infty$ and ε_t^2 is covariance stationary.*

Proof. See Appendix C.

Using the same steps presented in the Appendix C, we can demonstrate the existence of higher moments of ε_t^2 as follows.

Lemma 1. *Suppose that $\phi_{n,i}$s, $\beta_{n,j}$s and d_n satisfy the non-negativity constraints in Conrad and Haag (2006), $p_{ij} \in (0,1)$ for all $i, j \in \{1, 2, \ldots, N\}$, $E(\eta_t^{2m^*}) < \infty$ for all $m^* \leq m$ and $\sum_{k=1}^{\infty} \left[\sup_n b_{n,k} \right] < \left[E(\eta_t^{2m}) \right]^{-1/m}$, then finite $E(\varepsilon_t^{2m})$ exists and will not depend on t for $1 \leq m < \infty$.*

Proof. See Appendix D.

From the above results, it can be seen that the amplitude parameters γ_ns are crucial to determining the existence of the unique solution, weakly stationarity and finite higher moments of the MRS-HGARCH process. Since $\gamma_n = \sum_{k=1}^{\infty} b_{n,k}$, if any $\gamma_n \geq 1$, it is straightforward to verify that $\sum_{k=1}^{\infty} \left[\sup_n b_{n,k} \right] \geq 1$. Hence, the assumptions of the Theorems 1 and 2 and Lemma 2 would be violated in such a case. Therefore, $\gamma_n < 1$ for all $n = 1, 2, \ldots, N$ is a necessary condition for an MRS-HGARCH process to be weakly stationary and to have a unique solution with finite higher moments. This result is analogous to that proposed by Li et al. (2015).

In terms of the estimation, we can use the same procedure as for the MRS-GARCH model described in Section 3 to perform the MLE. Regarding the asymptotic theory for ML estimators of a general MRS-ARCH(∞) model, which nests the MRS-HGARCH, the literature is not complete yet, but some results can be emphasized. In particular, for

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the MRS-ARCH(p) model, where p is a known positive integer, Francq et al. (2001) show the consistency of the ML estimators. Moreover, under general regularity conditions, the seminal work of Douc et al. (2004) demonstrate a complete asymptotic normality distribution of the ML estimators for the MRS-AR(p)-type model, which nests the MRS-ARCH(p) as a special case. As we proxy the infinity by 1000 in Equation (2) during estimation, the MRS-ARCH(∞) process defined by Equation (11) is then estimated as an MRS-ARCH(1000) process. Therefore, if the assumptions 1–3 of Li et al. (2015) hold[9] and $p_{jk} \in (0,1)$ for all j and $k = 1, 2, \ldots, N$, the assumptions (A1)–(A8) of Douc et al. (2004) will hold for $\{\varepsilon_t^2\}$ in a general case. Also, $H(\theta_0)$ will be positive definite, where θ_0 is the true parameters vector and $H(\cdot)$ is the Fisher information of the MRS-HGARCH model (proxied by an MRS-ARCH(1000) process). Hence, using Theorems 4 and 6 of Douc et al. (2004)[10], we further have that

$$\sqrt{T}(\hat{\theta} - \theta_0) \overset{A}{\sim} N(0, H(\theta_0)^{-1}) \tag{15}$$

where $\hat{\theta}$ is the ML estimators vector and A stands for the asymptotic convergence.

To examine the existence of the hyperbolic memory in the conditional variance, we can test H_0: $d = 1$ against H_1: $d < 1$ for the MRS-HGARCH model. Since when $d = 1$, the MRS-HGARCH just reduces to a general MRS-GARCH model with no unidentified parameters, it is convenient to perform the standard likelihood ratio test (LRT) for this aim. If the assumptions of Equation (15) and Theorem 2 all hold, then the test statistic of the LRT (i.e. $-2(logL_{MG} - logL_{MH})$) would asymptotically follow a Chi-square distribution with 1 degree of freedom, where $logL_{MG}$ ($logL_{MH}$) is the logarithm of the likelihood of the fitted MRS-GARCH (MRS-HGARCH) model. It is worth noticing that in H_0, $d = 1$ lies exactly on the boundary. For the original HGARCH model, the score test investigated in Li et al. (2011) is argued to be more appropriate in this case. However, it is not straightforward to extend their approach to the MRS-HGARCH model. This is due to the complex structure of its likelihood, which contains

[9]Note that for Assumption 1 to be valid, in our case, it is further required that $\phi_n(X)$ and $\beta_n(X)$ polynomials have no common roots, and coefficients $\phi_{n,i}$s, $\beta_{n,j}$s and d_n satisfy the non-negativity constraints proposed by Conrad and Haag (2006). For Assumption 2, d_{min} is the lower bound of all d_n. Also, instead of the existence of a strictly stationary and ergodic solution of r_t, we require $\sum_{k=1}^{\infty} \left[\sup_n b_{n,k}\right] < 1$. For Assumption 3, true values of d_n all fall in the range (0.5,1) and $\kappa_4 \in (4/(2d_{0,min} + 3), 1)$, where $d_{0,min}$ is the minimum of true values of all d_n. In all cases, $n = 1, 2, \ldots, N$.

[10]In their paper, Theorem 4 proves the asymptotic consistency and normality of the ML estimator for a strictly stationary MRS-AR(p) process. Theorem 6 further proves those properties for a non-stationary case.

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an iterative filtering process shown in Equations (7)–(9). To the best of our knowledge, the literature on a powerful and applicable score test for the MRS-type model is not complete yet. A deeper examination of the score test is also beyond the scope of this paper. More importantly, when the assumptions of Equation (15) and Theorem 2 all hold, the classic trinity tests including the Wald test, LRT and score test would be asymptotically equivalent for our aim (Engle, 1984). Hence, the LRT is performed to study the significance of hyperbolic memory.

We hereby argue that the MRS-HGARCH framework can effectively distinguish between the pure HGARCH and pure MRS-GARCH DGPs. To verify this, we consider three scenarios of the MRS-HGARCH model: only ω is allowed to switch (Scenario 1); ϕ_is, β_js and γ are further allowed to switch (Scenario 2); and no restrictions (Scenario 3). The previously simulated data are fitted into the MRS-HGARCH models under each scenario, and the estimates are summarized in Tables 2, 3 and 4, respectively.

Comparing the results in Tables 1 and 2, similar to the HGARCH model, the restricted MRS-HGARCH model leads to consistent estimates of d (means are close to the true values) when the true DGP is HGARCH. This is also true for the MRS-GARCH DGP, where mean estimates are close to the true values 1. In terms of the significance, results in Table 2 suggest significant hyperbolic memory for the HGARCH DGP in almost all cases. For the MRS-GARCH DGP, however, most results show that the hypothesis of geometric memory cannot be rejected. Hence, the pure HGARCH and pure MRS-GARCH DGPs can be effectively distinguished. The same conclusions can be drawn from Table 3, where the short-memory and amplitude parameters are further allowed to switch between states.

Our results under Scenarios 1 and 2 demonstrate the effectiveness of the MRS-HGARCH model, when d is held constant. This is consistent with Shi and Ho (2015) focusing on the first-moment financial data. Moreover, Haldrup and Nielsen (2006) and Tsay and Härdle (2009) point out the importance that the behaviour of d could also be state-dependent. Therefore, we consider Scenario 3, where d is further allowed to switch between states. In Table 4, although the SEs are slightly greater than their counterparts in Tables 2 and 3, the estimates of both d_1 and d_2 still lead to the same conclusions: (in)significant hyperbolic memories are found in both states for HGARCH (MRS-GARCH) DGPs. Also, for the HGARCH DGP, the means of both estimated d_1 and d_2 are close to the true values of d, despite that the true DGP only has one regime. Hence, we demonstrate the effectiveness and robustness of our model to address the

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Table 2
Summary of MRS-HGARCH(1,*d*,1) Models: Scenario 1

(a) Simulated MRS-GARCH(1,1) Data

p_{11}	p_{22}	P_1	P_2	$Mean_d$	SE_d	*F.R.*
0.99	0.999	0.7	0.9	0.9563	0.0618	0.0367
		0.8	0.8	0.9521	0.0697	0.0400
		0.9	0.7	0.9337	0.0929	0.0467
0.999	0.99	0.7	0.9	0.9429	0.0798	0.0367
		0.8	0.8	0.9592	0.0612	0.0333
		0.9	0.7	0.9538	0.0660	0.0300
0.99	0.99	0.7	0.9	0.9641	0.0513	0.0267
		0.8	0.8	0.9418	0.0806	0.0433
		0.9	0.7	0.9312	0.0933	0.0467
0.999	0.999	0.7	0.9	0.9388	0.0766	0.0467
		0.8	0.8	0.9394	0.0781	0.0400
		0.9	0.7	0.9489	0.0677	0.0333

(b) Simulated HGARCH(1,d,1) Data

d	ϕ	β	$Mean_d$	SE_d	*F.R.*
0.45	0.10	0.20	0.4638	0.1254	1.0000
	0.15	0.15	0.4792	0.1323	1.0000
	0.20	0.10	0.4788	0.1362	1.0000
0.55	0.10	0.20	0.5708	0.1301	1.0000
	0.15	0.15	0.5611	0.1353	1.0000
	0.20	0.10	0.5573	0.1243	1.0000
0.60	0.10	0.20	0.6144	0.1293	0.9933
	0.15	0.15	0.6100	0.1258	0.9967
	0.20	0.10	0.6013	0.1272	1.0000

Note: for explanations of variables, please see Table 1.

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Table 3
Summary of MRS-HGARCH(1,*d*,1) Models: Scenario 2

(a) Simulated MRS-GARCH(1,1) Data

p_{11}	p_{22}	P_1	P_2	$Mean_d$	SE_d	*F.R.*
0.99	0.999	0.7	0.9	0.9576	0.0614	0.0333
		0.8	0.8	0.9527	0.0686	0.0367
		0.9	0.7	0.9365	0.0908	0.0433
0.999	0.99	0.7	0.9	0.9445	0.0782	0.0400
		0.8	0.8	0.9590	0.0589	0.0333
		0.9	0.7	0.9567	0.0622	0.0367
0.99	0.99	0.7	0.9	0.9596	0.0632	0.0300
		0.8	0.8	0.9429	0.0798	0.0433
		0.9	0.7	0.9364	0.0898	0.0467
0.999	0.999	0.7	0.9	0.9452	0.0717	0.0433
		0.8	0.8	0.9414	0.0756	0.0433
		0.9	0.7	0.9546	0.0631	0.0367

(b) Simulated HGARCH(1,d,1) Data

d	ϕ	β	$Mean_d$	SE_d	*F.R.*
0.45	0.10	0.20	0.4721	0.1364	1.0000
	0.15	0.15	0.4801	0.1391	1.0000
	0.20	0.10	0.4779	0.1376	1.0000
0.55	0.10	0.20	0.5750	0.1352	1.0000
	0.15	0.15	0.5747	0.1355	1.0000
	0.20	0.10	0.5693	0.1273	1.0000
0.60	0.10	0.20	0.6178	0.1354	0.9833
	0.15	0.15	0.6122	0.1281	0.9867
	0.20	0.10	0.6022	0.1280	0.9967

Note: for explanations of variables, please see Table 1.

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Table 4
Summary of MRS-HGARCH(1,d,1) Models: Scenario 3

(a) Simulated MRS-GARCH(1,1) Data

p_{11}	p_{22}	P_1	P_2	$Mean_{d_1}$	SE_{d_1}	$Mean_{d_2}$	SE_{d_2}	$F.R._1$	$F.R._2$
0.99	0.999	0.7	0.9	0.9502	0.0857	0.9670	0.0645	0.0433	0.0300
		0.8	0.8	0.9465	0.1053	0.9599	0.0805	0.0433	0.0367
		0.9	0.7	0.9301	0.1300	0.9415	0.1173	0.0500	0.0400
0.999	0.99	0.7	0.9	0.9431	0.1156	0.9483	0.1292	0.0467	0.0433
		0.8	0.8	0.9559	0.0594	0.9612	0.0878	0.0367	0.0300
		0.9	0.7	0.9543	0.0703	0.9606	0.0712	0.0367	0.0333
0.99	0.99	0.7	0.9	0.9580	0.0719	0.9612	0.0881	0.0333	0.0300
		0.8	0.8	0.9410	0.0880	0.9456	0.0957	0.0433	0.0400
		0.9	0.7	0.9304	0.1159	0.9384	0.1236	0.0500	0.0467
0.999	0.999	0.7	0.9	0.9412	0.0826	0.9501	0.0988	0.0433	0.0367
		0.8	0.8	0.9370	0.0952	0.9454	0.0760	0.0467	0.0400
		0.9	0.7	0.9507	0.0785	0.9589	0.0840	0.0367	0.0333

(b) Simulated HGARCH(1,d,1) Data

d	ϕ	β	$Mean_{d_1}$	SE_{d_1}	$Mean_{d_2}$	SE_{d_2}	$F.R._1$	$F.R._2$
0.45	0.10	0.20	0.4698	0.1401	0.4741	0.1326	1.0000	1.0000
	0.15	0.15	0.4774	0.1419	0.4819	0.1361	1.0000	1.0000
	0.20	0.10	0.4746	0.1380	0.4791	0.1335	1.0000	1.0000
0.55	0.10	0.20	0.5712	0.1373	0.5790	0.1367	1.0000	1.0000
	0.15	0.15	0.5739	0.1403	0.5781	0.1425	1.0000	1.0000
	0.20	0.10	0.5659	0.1368	0.5709	0.1420	1.0000	1.0000
0.60	0.10	0.20	0.6119	0.1359	0.6189	0.1322	0.9833	0.9800
	0.15	0.15	0.6102	0.1358	0.6166	0.1397	0.9867	0.9833
	0.20	0.10	0.6004	0.1323	0.6065	0.1370	0.9967	0.9933

Note: for explanations of variables, please see Table 1.

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spurious hyperbolic memory caused by regime switching in the second moment.

Finally, it is interesting to see how the MRS-HGARCH model performs when the true DGP contains both hyperbolic memory and regime switching. A two-state MRS-HGARCH(1,d,1) DGP is employed to generate the simulation, where μ, ω_1, ω_2, ϕ_1, β_1, ϕ_2, β_2, γ_1 and γ_2 are set to 0, 0.1, 0.5, 0.15, 0.15, 0.15, 0.15, 0.4 and 0.6, respectively. Additionally, the Student's t-distributed innovation is used with $v = 3$. There are twelve combinations of different p_{11}, p_{22}, d_1 and d_2, the specific values of which are consistent with those in previous simulations. We still generate 300 replicates with 5,000 simulations for each combination. To avoid the starting bias, we adopt the same strategies described in Section 4. Each simulated dataset is then fitted into the HGARCH and MRS-HGARCH models, respectively, and the estimates are summarized in Table 5.

Table 5
Summary of Simulated MRS-HGARCH(1,d,1) Data

p_{11}	p_{22}	d_1	d_2	$Mean_{d_1}$	SE_{d_1}	$Mean_{d_2}$	SE_{d_2}	$F.R._1$	$F.R._2$
(a) HGARCH Model									
0.99	0.999	0.45	0.60	0.5222	0.1336			1.0000	
		0.55	0.55	0.4860	0.1401			1.0000	
		0.60	0.45	0.4925	0.1353			1.0000	
0.999	0.99	0.45	0.60	0.5212	0.1519			1.0000	
		0.55	0.55	0.4922	0.1553			1.0000	
		0.60	0.45	0.5048	0.1525			1.0000	
0.99	0.99	0.45	0.60	0.5534	0.1352			1.0000	
		0.55	0.55	0.4731	0.1331			1.0000	
		0.60	0.45	0.5261	0.1347			1.0000	
0.999	0.999	0.45	0.60	0.4498	0.1284			1.0000	
		0.55	0.55	0.4520	0.1271			1.0000	
		0.60	0.45	0.4794	0.1257			1.0000	
(b) MRS-HGARCH Model									
0.99	0.999	0.45	0.60	0.4574	0.1016	0.6061	0.1075	1.0000	0.9933
		0.55	0.55	0.5682	0.0987	0.5666	0.1046	1.0000	1.0000
		0.60	0.45	0.5917	0.0917	0.4412	0.0990	1.0000	1.0000
0.999	0.99	0.45	0.60	0.4459	0.1283	0.5920	0.1301	1.0000	1.0000
		0.55	0.55	0.5514	0.1253	0.5474	0.1304	1.0000	1.0000
		0.60	0.45	0.6024	0.1260	0.4581	0.1232	0.9967	1.0000
0.99	0.99	0.45	0.60	0.4419	0.0938	0.6034	0.0947	1.0000	0.9933
		0.55	0.55	0.5685	0.1115	0.5709	0.1019	1.0000	1.0000
		0.60	0.45	0.6019	0.0970	0.4512	0.0932	0.9967	1.0000
0.999	0.999	0.45	0.60	0.4561	0.0963	0.6061	0.0911	1.0000	0.9900
		0.55	0.55	0.5597	0.0948	0.5582	0.0893	1.0000	1.0000
		0.60	0.45	0.6021	0.0930	0.4564	0.0886	0.9967	1.0000

Note: for explanations of variables, please see Table 1.

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The estimates of d from the HGARCH model in Table 5 are all significantly smaller than 1. For the cases $d_1 \neq d_2$, the mean estimates of d mostly fall between the true values of 0.45 and 0.60. Also, most of them tend to be closer to 0.45. As demonstrated above, the MRS-HGARCH model can properly control for the effect of the regime switching and should provide more reliable estimators of the hyperbolic-memory parameters. In Table 5, it can be seen that the estimates of d_1 and d_2 are almost all significantly smaller than 1, and their means are close to the true values in all cases. Besides, the obtained SEs in the MRS-HGARCH model are mostly smaller than those in the HGARCH model. Hence, the MRS-HGARCH model outperforms the HGARCH under various circumstances. Additionally, for other essential parameters like the transition probabilities (not reported and available upon request), the mean estimates of the MRS-HGARCH model are all close to their true values with small SEs. Thus, it suggests that the MRS-HGARCH model can correctly identify the underlying volatility states of the true DGP.

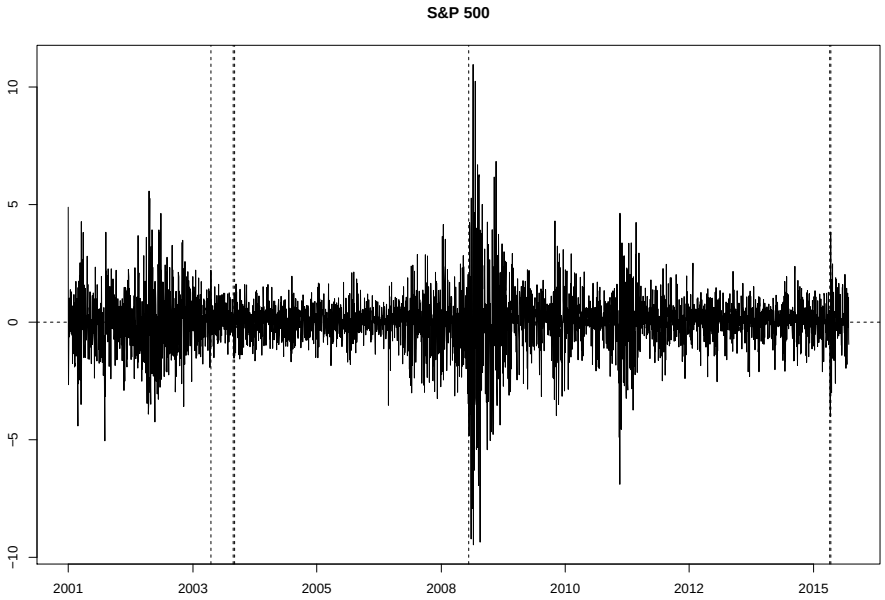
In conclusion, via systematically designed Monte Carlo studies, we have demonstrated that the proposed MRS-HGARCH model can effectively distinguish between the pure HGARCH and pure MRS-GARCH DGPs. More specifically, if the estimated d is significantly smaller than 1 in the HGARCH model but is insignificant in all states of the MRS-HGARCH model, we can conclude that the hyperbolic memory is purely caused by regime switching. On the other hand, if the estimated d_{s_t} in the MRS-HGARCH model is significant, we can safely conclude that the hyperbolic memory in state s_t is not spurious. In addition, when the true DGP contains both hyperbolic memory and regime switching, the MRS-HGARCH model can generate consistent estimates of the hyperbolic-memory persistences and effectively identify the structure of the underlying volatility states.

6. Empirical results

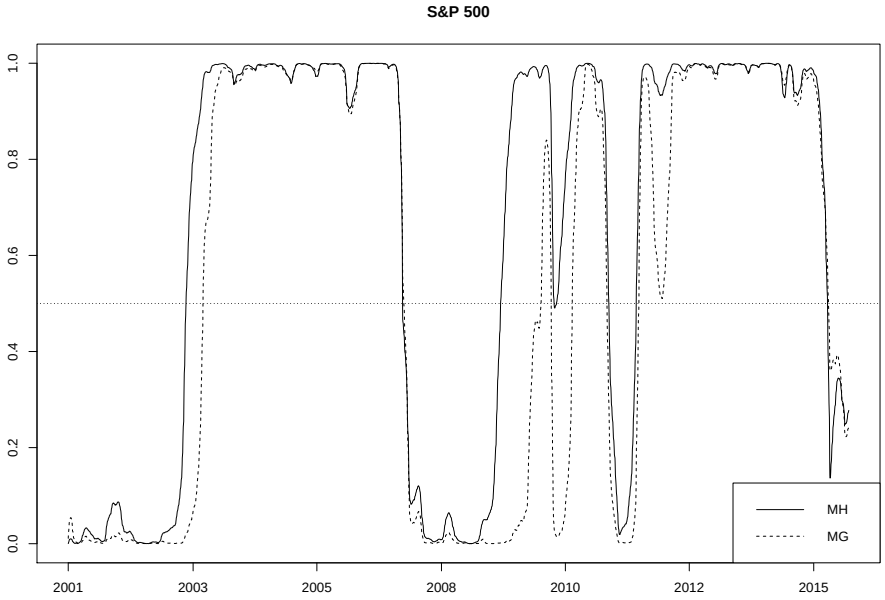
To empirically compare our MRS-HGARCH framework with the HGARCH and MRS-GARCH models, we fit them for the daily returns of the S&P 500 index. The daily closing prices for S&P 500 index over the period 1 January 2001 to 31 December 2015 are obtained from the Thomson Reuters Tick History (TRTH database). The corresponding return in the percentage series is defined as the logarithm of the daily closing price differences times 100; that is, $r_t = 100 \times \log(S_t/S_{t-1})$.

The daily return series of the S&P 500 index are plotted in Figure 1 (a). It can be seen

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(a) Filtered Returns



(b) Smoothing Probability of Calm State

Fig. 1. Filtered Returns and Smoothing Probability of Calm State of the S&P 500 Index

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that the return is relatively more volatile between 2001 and 2003, and then it becomes more stable. From 2008 to 2010, the return is much more volatile again. After 2010, the volatility tends to be smaller, with some turbulences around the end of 2010, the start of 2012 and the second half year of 2015. In addition, the mean and standard error of the S&P 500 return are 0.0123 and 1.2566, respectively. The skewness is -0.1994, indicating that the S&P 500 returns are roughly symmetric. The kurtosis is 8.6664, suggesting a non-Gaussian distribution. Thus, we perform the Kolmogorov–Smirnov and Jarque–Bera normality tests on the standardized residuals of the fitted GARCH(1,1) model with Gaussian distribution (not presented and available upon request). The null hypotheses indicating normality are rejected in both cases. To test the existence of structural changes in volatility, we employ the non-parametric change point detection algorithm proposed by Ross (2013), which is suitable for non-Gaussian distributed data. The detected locations of changes are plotted in Figure 1 (a) as vertical dotted lines. The multiple changes indicate that a spurious hyperbolic memory might be present. We hereby fit the S&P 500 data into the HGARCH, MRS-GARCH and MRS-HGARCH models, all with a Student's t-distribution.[11]

The estimates are presented in Table 6. The estimated d and γ in the HGARCH model are 0.7209 and 0.9269, showing that the conditional variance is stationary and the unconditional variance exists (Li et al., 2015). v is estimated to be significant and around 7.5, indicating a significant non-Gaussian distribution. Turning to the MRS-GARCH model, estimates of p_{11} and p_{22} are around 0.997. This suggests a considerably small frequency for the returns to switch between volatility states (Ho et al., 2013). In addition, $\alpha_1 + \beta_1$ is around 0.9065, while $\alpha_2 + \beta_2$ is 0.9838. This indicates that the volatility persistences in the calm and turbulent states are both quite large. Besides, the estimate of v is around 8.5. In terms of model performance evaluations, logarithm of likelihood and Bayesian Information Criterion (BIC) both demonstrate that the HGARCH model outperforms the MRS-GARCH model. The common residual diagnostics including the Brock-Dechert-Scheinkman (BDS) test, Ljung-Box test (for the squared standardized residuals) and ARCH LM test suggest that both models are adequate.

As demonstrated in Section 5, the MRS-HGARCH model can properly control for the effect of regime switching and provide more reliable estimates of d. In Table 6, the

[11]For reasons discussed in Sections 3 and 4, we only present the GARCH(1,1) specification and two-state case for all relevant models here. We have also considered the GARCH(1,2), GARCH(2,1) and GARCH(2,2) with three- and four-state assumptions, the conclusions are consistent with those discussed in this paper.

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Table 6
Summary of Model Fitting Results: S&P 500 Daily Returns

	HGARCH		MRS-GARCH		MRS-HGARCH
ω	0.0379	ω_1	0.0482	ω_1	0.1264
	(0.0099)		(0.0147)		(0.0285)
		ω_2	0.0505	ω_2	0.0246
			(0.0200)		(0.0093)
ϕ	-0.0263	α_1	0.0881	ϕ_1	-0.1317
	(0.0410)		(0.0195)		(0.1548)
β	0.6859	β_1	0.8184	β_1	0.4638
	(0.0666)		(0.0400)		(0.0472)
		α_2	0.0938	ϕ_2	-0.0299
			(0.0160)		(0.0418)
		β_2	0.8900	β_2	0.7506
			(0.0173)		(0.0568)
γ	0.9269			γ_1	0.6595
	(0.0471)				(0.0126)
				γ_2	0.9823
					(0.0156)
		p_{11}	0.9971	p_{11}	0.9978
			(0.0017)		(0.0020)
		p_{22}	0.9969	p_{22}	0.9965
			(0.0018)		(0.0012)
v	7.5041	v	8.4809	v	7.8460
	(0.4440)		(0.5499)		(0.3804)
d	0.7209			d_1	0.8551
	(0.0854)				(0.0934)
				d_2	0.7715
					(0.0858)
BDS_{10}	-0.2859	BDS_{10}	-0.5708	BDS_{10}	-0.3163
Q^2_{10}	0.3559	Q^2_{10}	0.5285	Q^2_{10}	0.3008
$ARCH_{10}$	0.3668	$ARCH_{10}$	0.5331	$ARCH_{10}$	0.3161
$E(\lvert\hat{\eta}_t\rvert)$	0.7555	$E(\lvert\hat{\eta}_t\rvert)$	0.7537	$E(\lvert\hat{\eta}_t\rvert)$	0.7530
$Var(\hat{\eta}_t)$	0.9940	$Var(\hat{\eta}_t)$	0.9709	$Var(\hat{\eta}_t)$	0.9798
$Skew.$	-0.4227	$Skew.$	-0.3217	$Skew.$	-0.4251
$Kurt.$	1.3795	$Kurt.$	0.8310	$Kurt.$	1.2820
$log.lik$	-5238	$log.lik$	-5246	$log.lik$	-5223
BIC	10571	BIC	10583	BIC	10534
In	0.1226	In	0.1275	In	0.1128
Out_1	0.2638	Out_1	0.2706	Out_1	0.2155
Out_{10}	0.8913	Out_{10}	1.0218	Out_{10}	0.7712

Note: $E(|\hat{\eta}_t|)$, $Var(\hat{\eta}_t)$, $Skew.$ and $Kurt.$ are the sample expectation of the absolute standardized residuals, sample variance, skewness and kurtosis of the standardized residuals, respectively. BDS_{10} is the Brock-Dechert-Scheinkman test statistics of 2 times stand error of the standardized residuals at embedding dimension 10. Q^2_{10} is the Ljung-Box test statistics for the squared standardized residuals at lag 10. $ARCH_{10}$ is the ARCH LM test statistics at lag 10. $log.lik$ is the logarithm of likelihood. BIC is the Bayesian Information Criterion. Values in the parentheses are the corresponding standard errors. In is the average in-sample forecasting error. Out_1 and Out_{10} are the average out-of-sample error of the one- and ten-step-ahead forecasts over the 100 rolling-window sample, respectively.

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estimated d_1 and d_2 from the MRS-HGARCH model are 0.8551 and 0.7715, respectively. Both of them are greater than the estimated d in the HGARCH model. Moreover, according to the LRT, d_2 (d_1) is (not) significantly smaller than 1 at 5% level. Hence, it is suggested that the conditional variance only exhibits significant hyperbolic memory in the turbulent state. Further, it supports our argument that after controlling for the regime-switching effect, the hyperbolic-memory persistences should be weaker. Nevertheless, significant hyperbolic memory is still observed even when the MRS-HGARCH model is employed. It draws attention to the necessity and importance to consider the effects of hyperbolic memory to study the financial volatility in practice.

In terms of the MRS properties, the MRS-HGARCH model generates similar estimates of p_{11} and p_{22} to those of the MRS-GARCH. This may indicate that both models may identify similar underlying state-series s_t. To examine this, the estimated smoothing probability series $P(s_t = 1|\Omega_T)$ from the MRS-GARCH (MG) and MRS-HGARCH (MH) models are plotted in Figure 1 (b). Despite that $P(s_t = 1|\Omega_T)$ of MG indicates longer turbulent periods, the patterns of both series are close to each other. Moreover, those patterns are consistent with our observation described above: between 2001 and 2003, the S&P 500 returns lie in the turbulent state; between 2004 and 2008, they mostly fall in the calm state, with some short-term switches to the turbulent state; from 2008 to 2010, the return series tend to switch back to the turbulent state and stay there; and after 2010, they mostly lie in the calm state except for some short-term turbulences. It also complies with the real macroeconomic situation: between 2001 and 2003, the macro-economy suffers from the effects of the burst of Dot-com bubble; and the Global Financial Crisis (GFC) affects the macro-economy from 2008 to 2010.

In addition, the estimated v is not far from those of HGARCH and MRS-GARCH and is around 7.85. This indicates that the distributional features of the corresponding innovation sequences of all models should be similar to each other (Haas et al., 2010). To verify this, we calculate the standardized residuals for each model ($\hat{\eta}_t = (r_t - \hat{\mu})/\sqrt{\hat{h}_t}$).[12] As expected, their sample statistics across each model are relatively close. Those statistics include expectation of absolute value, variance, skewness and kurtosis, as adopted in Haas et al. (2010). Nevertheless, the residual diagnostics suggest that the MRS-HGARCH model is also adequate.

More interestingly, the logarithm of likelihood and BIC show that the MRS-

[12] According to Haas et al. (2010), $\hat{\eta}_t$ generated by an MRS-type model is essentially the average of state-specific standard residuals weighted by update probabilities $P(s_t|\Omega_{t-1})$ and $E(\hat{\eta}_t^2)$ (1 in our case).

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HGARCH model outperforms both the HGARCH and MRS-GARCH models. To consider the in-sample forecasting performance, we measure the prediction error for each model as the average $\left|\sqrt{\hat{h}_t} - RV_t\right|$. $\hat{h}_t$ is the fitted h_t for each model, and RV_t is the realized volatility constructed by intra-day hourly returns to proxy the conditional volatility. Regarding the out-of-sample forecasts, we use the last 100 observations as the prediction sample and the others as the training sample. Then, we fit each model using the training sample and calculate the one- and ten-step-ahead forecasts of h_t, respectively. After that, we include the first observation of the prediction sample in the training sample and generate another one- and ten-step-ahead forecasts. We repeat this rolling-window approach until the 100 one- and ten-step-ahead forecasts are produced. Finally, we calculate the average absolute difference between the forecasts and the corresponding RV_t, which measures the out-of-sample forecasting errors. According to the produced average errors, the MRS-HGARCH model consistently beats the other two for both the in- and out-of-sample forecasting performance.

In conclusion, using the empirical results of the daily returns of the S&P 500, we demonstrate that the MRS-HGARCH model can effectively identify the structure of the underlying volatility states, which is consistent with the real trend of the macro-economy. Compared with the HGARCH model, it can generate state-dependent and potentially more reliable estimates of the hyperbolic-memory persistences. In terms of model evaluations and forecasting performance, MRS-HGARCH beats both the HGARCH and MRS-GARCH models. Nevertheless, All of them demonstrate acceptable residual diagnostics results. Thus, MRS-HGARCH outperforms both the HGARCH and MRS-GARCH models and could be a widely useful tool for modelling financial volatility in other contexts.

7. Concluding remarks

Diebold and Inoue (2001) argue that the long memory and regime switching are easily confused with each other and should be studied as interchangeable concepts. However, recent studies like Perron and Qu (2010), Qu (2011) and Shi and Ho (2015) argue that if the cause of this confusion is properly controlled, the two processes can be distinguished for the first-moment data. Since covariance stationary ARCH(∞) processes cannot have true long memory as its first-moment counterparts, we use the concept of hyperbolic memory as suggested by Davidson (2004). Based on the existing literature, we propose an effective approach to distinguish between hyperbolic memory and regime switching

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for the second-moment financial series. The HGARCH and MRS-GARCH models are employed to estimate the hyperbolic memory and regime switching of the conditional variance of financial series, respectively. Also, as financial series are rarely Gaussian, we employ the Student's t-distribution in this paper.

First, we derive the theoretical cause of the confusion between the hyperbolic memory and regime switching in the second moment. We then perform a simulation study to demonstrate that the MRS-GARCH DGP can lead to significant estimates of hyperbolic memory. This is consistent with the results of Mikosch and Starica (2003) and Lai and Xing (2006), who suggest that structural changes can lead to spurious hyperbolic memory in the second moment.

Following the idea of Shi and Ho (2015), we then incorporate the regime switching into the HGARCH framework and propose the MRS-HGARCH model. Using seminal works of Francq et al. (2001), Douc et al. (2004), Robinson et al. (2006) and Li et al. (2015), we study the statistical properties including the existence of the unique solution, weakly stationarity and unconditional higher moments of the MRS-HGARCH process. The asymptotic properties of the ML estimators are also discussed, assuming that the MRS-ARCH(∞) is proxied by an MRS-ARCH(1000) process in practice. We further demonstrate that the MRS-HGARCH model can effectively distinguish between the pure HGARCH and pure MRS-GARCH DGPs via simulations under various scenarios. In particular, if the estimated d_{s_t} in the MRS-HGARCH model is significantly smaller than 1, we can safely conclude that the hyperbolic memory in state s_t is not spurious. Additionally, when the true DGP contains both hyperbolic memory and regime switching, the MRS-HGARCH framework is capable of identifying the underlying volatility states and providing consistent and more reliable estimators of the hyperbolic-memory persistences.

We conduct an empirical study for the daily returns of the S&P 500 index to compare the performance among the HGARCH, MRS-GARCH and MRS-HGARCH models. It is demonstrated that the MRS-HGARCH model can effectively identify the underlying volatility states, the structure of which is consistent with the real trend of the macroeconomy. Compared with the HGARCH model, it can generate state-dependent and potentially more reliable estimates of the hyperbolic-memory persistence. Additionally, significant hyperbolic memory is still observed after controlling for the regime switching in the high-volatility state. It draws attention to the necessity and importance to consider the effects of hyperbolic memory when studying the financial volatility in prac-

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tice. In terms of model evaluations and forecasting performance, the MRS-HGARCH framework beats both the HGARCH and MRS-GARCH models. Nevertheless, all the models demonstrate acceptable residual diagnostics results. Therefore, MRS-HGARCH can be a widely useful tool for modelling the hyperbolic-memory characteristic of the second-moment series in other contexts.

Our paper also sheds some lights on the possible directions of future works. First, the parameters of the unconditional mean and other innovation distribution can be further allowed to be regime-switching. A thorough discussion on the statistical properties of such a more flexible process is of interest in practice to study the first- and second-moment behaviours simultaneously. Second, the overall volatility persistence of the MRS-GARCH process is derived in Haas et al. (2004). Due to the significant differences in the memory structure, however, their theoretical result cannot be straightforwardly extended to the MRS-HGARCH case. A form investigation on deriving its overall hyperbolic memory measurement remains for the future work. Additionally, our paper demonstrates the effectiveness of the MRS-HGARCH model to consistently estimate the hyperbolic memory and regime switching impacts. Decomposing contributions of those impacts to the conditional variance sequence is out of the scope of this paper and is open to the future research.

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Appendices

A. Proof of Proposition 1

$$
\begin{aligned}
&var(\sum_{t=1}^{T} e_t^2) = var(\sum_{t=1}^{T}[I(s_t=1)\varepsilon_{1,t}^2 + I(s_t=2)\varepsilon_{2,t}^2]) \\
=&var(\sum_{t=1}^{T} I(s_t=1)\varepsilon_{1,t}^2) + var(\sum_{t=1}^{T} I(s_t=2)\varepsilon_{2,t}^2) + \\
&2\sum_{i=1}^{T}\sum_{j=1}^{T} cov(I(s_i=1)\varepsilon_{1,i}^2,\, I(s_j=2)\varepsilon_{2,j}^2)
\end{aligned}
$$

Next, we will derive the first variance term, as the second one will have a similar structure. Since $\varepsilon_{1,t}$ is a well-defined GARCH(1,1) process and $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$, using the results of Bollerslev (1988), $\varepsilon_{1,t}^2$ is covariance stationary. Using the proof of Proposition 3 in Diebold and Inoue (2001), we have that

$$
\begin{aligned}
&var(\sum_{t=1}^{T} I(s_t=1)\varepsilon_{1,t}^2) \\
=&\sum_{t=1}^{T} var(I(s_t=1)\varepsilon_{1,t}^2) + 2\sum_{j=1}^{T-1}(T-j)cov(I(s_t=1)\varepsilon_{1,t}^2,\, I(s_{t-j}=1)\varepsilon_{1,t-j}^2)
\end{aligned}
$$

and

$$
\begin{aligned}
&cov(I(s_t=1)\varepsilon_{1,t}^2,\, I(s_{t-j}=1)\varepsilon_{1,t-j}^2) \\
=&E(I(s_t=1)I(s_{t-j}=1)\varepsilon_{1,t}^2\varepsilon_{1,t-j}^2) - E(I(s_t=1)\varepsilon_{1,t}^2)E(I(s_{t-j}=1)\varepsilon_{1,t-j}^2) \\
=&E(I(s_t=1)I(s_{t-j}=1))E(\varepsilon_{1,t}^2\varepsilon_{1,t-j}^2) - E(I(s_t=1))E(I(s_{t-j}=1))E(\varepsilon_{1,t}^2)E(\varepsilon_{1,t-j}^2)
\end{aligned}
$$

We now define that $cov(\varepsilon_{1,t}^2,\, \varepsilon_{1,t-j}^2) = \Gamma_j$, i.e. Γ_j is the j-th autocovariance function of $\varepsilon_{1,t}^2$ and do not depend on t. Also, as s_t is irreducible, $E(I(s_t=1)) = P(s_t=1) = (1-p_{22})/(2-p_{11}-p_{22})$ and $E(I(s_t=1)I(s_{t-j}=1)) = \lambda^j var(I(s_t=1)) + P(s_t=1)^2$, where $\lambda = p_{11}+p_{22}-1$ and $var(I(s_t=1)) = (1-p_{11})(1-p_{22})/(2-p_{11}-p_{22})^2$ (Hamilton, 1994).

Hence,

$$
\begin{aligned}
&cov(I(s_t=1)\varepsilon_{1,t}^2,\, I(s_{t-j}=1)\varepsilon_{1,t-j}^2) \\
=&[\lambda^j var(I(s_t=1)) + P(s_t=1)^2][\Gamma_j + E(\varepsilon_{1,t}^2)^2] - P(s_t=1)^2E(\varepsilon_{1,t}^2)^2 \\
=&\lambda^j var(I(s_t=1))\Gamma_j + \lambda^j var(I(s_t=1))E(\varepsilon_{1,t}^2)^2 + P(s_t=1)^2\Gamma_j
\end{aligned}
$$

Therefore, we have that

$$
\begin{aligned}
&\sum_{j=1}^{T-1}(T-j)cov(I(s_t=1)\varepsilon_{1,t}^2,\, I(s_{t-j}=1)\varepsilon_{1,t-j}^2) \\
=&\sum_{j=1}^{T-1}(T-j)\lambda^j var(I(s_t=1))\Gamma_j + \sum_{j=1}^{T-1}(T-j)\lambda^j var(I(s_t=1))E(\varepsilon_{1,t}^2)^2 +
\end{aligned}
$$

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$$\sum_{j=1}^{T-1}(T-j)P(s_t=1)^2E(\varepsilon_{1,t}^2)^2$$

According to Bollerslev (1988), when $3\alpha_1^2+2\alpha_1\beta_1+\beta_1^2<1$, $\Gamma_j=(\alpha_1+\beta_1)^{j-1}\rho_1 var(\varepsilon_{1,t}^2)$, $\rho_1=[\alpha_1(1-\alpha_1\beta_1-\beta_1^2)]/(1-2\alpha_1\beta_1-\beta_1^2)$, $E(\varepsilon_{1,t}^2)=\omega_1/(1-\alpha_1-\beta_1)$ and $E(\varepsilon_{1,t}^4)=3\omega_1^2(1+\alpha_1+\beta_1)/[(1-\alpha_1-\beta_1)(1-\beta_1^2-2\alpha_1\beta_1-3\alpha_1^2)]$. Thus,

$$\begin{aligned}
&\sum_{j=1}^{T-1}(T-j)\lambda^j var(I(s_t=1))\Gamma_j\\
=&\sum_{j=1}^{T-1}(T-j)\lambda^j var(I(s_t=1))(\alpha_1+\beta_1)^{j-1}\rho_1 var(\varepsilon_{1,t}^2)\\
=&var(I(s_t=1))var(\varepsilon_{1,t}^2)\rho_1(\alpha_1+\beta_1)^{-1}\sum_{j=1}^{T-1}(T-j)\lambda^j(\alpha_1+\beta_1)^j
\end{aligned}$$

and

$$\sum_{j=1}^{T-1}(T-j)\lambda^j(\alpha_1+\beta_1)^j=\frac{T(\alpha_1+\beta_1)\lambda}{1-(\alpha_1+\beta_1)\lambda}-\frac{(\alpha_1+\beta_1)\lambda-[(\alpha_1+\beta_1)\lambda]^T}{[1-(\alpha_1+\beta_1)\lambda]^2}$$

Since λ, $\alpha_1+\beta_1\in(0,1)$ and $\lambda=p_{11}+p_{22}-1=1-c_1T^{-\delta_1}-c_2T^{-\delta_2}$, we have that $[1-(\alpha_1+\beta_1)\lambda]^{-1}=O(T^{min(\delta_1,\delta_2)})$. Hence,

$$\begin{aligned}
\frac{T(\alpha_1+\beta_1)\lambda}{1-(\alpha_1+\beta_1)\lambda}&=O(T)\\
\frac{(\alpha_1+\beta_1)\lambda-[(\alpha_1+\beta_1)\lambda]^T}{[1-(\alpha_1+\beta_1)\lambda]^2}&=O(\frac{1}{1-(\alpha_1+\beta_1)\lambda})=O(T^{min(\delta_1,\delta_2)})
\end{aligned}$$

and

$$\sum_{j=1}^{T-1}(T-j)\lambda^j(\alpha_1+\beta_1)^j=O(T^{max(1,min(\delta_1,\delta_2))})=O(T^{min(\delta_1,\delta_2)})$$

We now consider the multipliers of the summation $var(I(s_t=1))var(\varepsilon_{1,t}^2)\rho_1(\alpha_1+\beta_1)^{-1}$. Among them, only $var(I(s_t=1))$ is related to T and

$$\begin{aligned}
var(I(s_t=1))=&\frac{(1-p_{11})(1-p_{22})}{(2-p_{11}-p_{22})^2}\\
=&\frac{c_1c_2T^{-(\delta_1+\delta_2)}}{c_1^2T^{-2\delta_1}+2c_1c_2T^{-(\delta_1+\delta_2)}+c_2^2T^{-2\delta_2}}\\
=&O(T^{-|\delta_1-\delta_2|})
\end{aligned}$$

Combining the above results, we get that

$$var(I(s_t=1))var(\varepsilon_{1,t}^2)\rho_1(\alpha_1+\beta_1)^{-1}\sum_{j=1}^{T-1}(T-j)\lambda^j(\alpha_1+\beta_1)^j=O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})$$

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Similarly,

$$\sum_{j=1}^{T-1}(T-j)\lambda^j var(I(s_t=1))E(\varepsilon_{1,t}^2)^2 = O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})$$

We now consider $\sum_{j=1}^{T-1}(T-j)P(s_t=1)^2\Gamma_j$, where

$$P(s_t=1) = (c_1T^{-\delta_2})/(c_1T^{-\delta_1}+c_2T^{-\delta_2}) = O(T^{min(\delta_1,\delta_2)-\delta_2})$$

and

$$\begin{aligned}\sum_{j=1}^{T-1}(T-j)\Gamma_j =&\rho_1 var(\varepsilon_{1,t}^2)\sum_{j=1}^{T-1}(T-j)(\alpha_1+\beta_1)^{j-1}\\ =&\rho_1 var(\varepsilon_{1,t}^2)(\frac{T}{1-\alpha_1-\beta_1}-\frac{1-(\alpha_1+\beta_1)^{T-1}}{(1-\alpha_1-\beta_1)^2})\\ =&O(T)\end{aligned}$$

Hence, $\sum_{j=1}^{T-1}(T-j)P(s_t=1)^2\Gamma_j = O(T^{2min(\delta_1,\delta_2)-2\delta_2+1})$.

On the other hand,

$$\begin{aligned}var(I(s_t=1)\varepsilon_{1,t}^2) =&E(I(s_t=1)^2\varepsilon_{1,t}^4) - E(I(s_t=1)\varepsilon_{1,t}^2)^2\\ =&E(I(s_t=1)^2)E(\varepsilon_{1,t}^4) - E(I(s_t=1))^2E(\varepsilon_{1,t}^2)^2\\ =&P(s_t=1)E(\varepsilon_{1,t}^4) - P(s_t=1)^2E(\varepsilon_{1,t}^2)^2\end{aligned}$$

We have used the fact that $E(I(s_t=1)^2) = E(I(s_t=1)) = P(s_t=1)$ in the above equation. Further, as $E(\varepsilon_{1,t}^2)$ and $E(\varepsilon_{1,t}^4)$ exist and do not depend on T, and $0 < \alpha_1+\beta_1 < 1$, we have that

$$var(I(s_t=1)\varepsilon_{1,t}^2) = O(T^{min(\delta_1,\delta_2)-\delta_2}) + O(T^{2min(\delta_1,\delta_2)-2\delta_2}) = O(T^{min(\delta_1,\delta_2)-\delta_2})$$

Therefore,

$$\begin{aligned}&var(\sum_{t=1}^{T} I(s_t=1)\varepsilon_{1,t}^2)\\ =&\sum_{t=1}^{T} var(I(s_t=1)\varepsilon_{1,t}^2) + 2\sum_{j=1}^{T-1}(T-j)cov(I(s_t=1)\varepsilon_{1,t}^2,\, I(s_{t-j}=1)\varepsilon_{1,t-j}^2)\\ =&O(T^{min(\delta_1,\delta_2)-\delta_2+1}) + O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})\end{aligned}$$

Similarly, it can be shown that $P(s_t=2) = O(T^{min(\delta_1,\delta_2)-\delta_1})$ and

$$var(\sum_{t=1}^{T} I(s_t=1)\varepsilon_{1,t}^2) = O(T^{min(\delta_1,\delta_2)-\delta_1+1}) + O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})$$

Finally, we consider the covariance $2\sum_{i=1}^{T}\sum_{j=1}^{T} cov(I(s_i=1)\varepsilon_{1,i}^2,\, I(s_j=2)\varepsilon_{2,j}^2)$. As $\eta_{1,t}$, $\eta_{2,t}$, $h_{1,t}$ and $h_{2,t}$ are independent, $\varepsilon_{1,t} = \eta_{1,t}\sqrt{h_{1,t}}$ and $\varepsilon_{2,t} = \eta_{2,t}\sqrt{h_{2,t}}$ are

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independent of each other. Therefore, we have that

$$\begin{aligned}
&cov(I(s_i=1)\varepsilon_{1,i}^2,\, I(s_j=2)\varepsilon_{2,j}^2)\\
=&E(I(s_i=1)\varepsilon_{1,i}^2 I(s_j=2)\varepsilon_{2,j}^2) - E(I(s_i=1)\varepsilon_{1,i}^2)^2 E(I(s_j=2)\varepsilon_{2,j}^2)^2\\
=&E(I(s_i=1)I(s_j=2))E(\varepsilon_{1,i}^2)E(\varepsilon_{2,j}^2) - E(I(s_i=1))E(I(s_j=2))E(\varepsilon_{1,i}^2)E(\varepsilon_{2,j}^2)\\
=&E(\varepsilon_{1,i}^2)E(\varepsilon_{2,j}^2)cov(I(s_i=1),\, I(s_j=2))
\end{aligned}$$

where

$$cov(I(s_i=1),\, I(s_j=2)) = cov(I(s_i=1),\, 1-I(s_j=1)) = -cov(I(s_i=1),\, I(s_j=1))$$

Hence,

$$\begin{aligned}
&\sum_{i=1}^{T}\sum_{j=1}^{T} cov(I(s_i=1)\varepsilon_{1,i}^2,\, I(s_j=2)\varepsilon_{2,j}^2)\\
=&-E(\varepsilon_{1,i}^2)E(\varepsilon_{2,j}^2)[\sum_{t=1}^{T} var(I(s_t=1)) + 2\sum_{j=1}^{T-1}(T-j)cov(I(s_t=1),\, I(s_{t-j}=1)]\\
=&-E(\varepsilon_{1,i}^2)E(\varepsilon_{2,j}^2)[Tvar(I(s_t=1)) + 2\sum_{j=1}^{T-1}(T-j)\lambda^j var(I(s_t=1))]\\
=&O(T^{1-|\delta_1-\delta_2|}) + O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})\\
=&O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})
\end{aligned}$$

Eventually, combing the above results can lead to

$$\begin{aligned}
&var(\sum_{t=1}^{T} e_t^2)\\
=&var(\sum_{t=1}^{T} I(s_t=1)\varepsilon_{1,t}^2) + var(\sum_{t=1}^{T} I(s_t=2)\varepsilon_{2,t}^2) + 2\sum_{i=1}^{T}\sum_{j=1}^{T} cov(I(s_i=1)\varepsilon_{1,i}^2,\, I(s_j=2)\varepsilon_{2,j}^2)\\
=&O(T^{min(\delta_1,\delta_2)-\delta_1+1}) + O(T^{min(\delta_1,\delta_2)-\delta_2+1}) + O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})\\
=&O(T) + O(T^{min(\delta_1,\delta_2)-|\delta_1-\delta_2|})\\
=&O(T^{max(1,min(\delta_1,\delta_2)-|\delta_1-\delta_2|)})
\end{aligned}$$

B. Proof of Theorem 1

When the non-negativity conditions proposed by Conrad (2006) are followed, $a_n(t)$ for all n and t will be non-negative. Further using the assumption that $\sum_{k=1}^{\infty}\left[\sup_n b_{n,k}\right] < 1$, the conditions (3) and (4) of the Assumption 1 of Dahlhaus et al. (2006) will be jointly satisfied. As $\sum_{n=1}^{N}[I(s_t=n)] = 1$ for all t, $|a_j(u) - a_j(v)| \leq \left[\sup_n b_{n,j}\right] - \left[\inf_n b_n, j\right] < 1$ for all u and v. Hence, we can direct apply the proof of the Proposition 1 in Dahlhaus et al. (2006) to prove that one solution of ε_t exists with h_t having the following form.

$$h_t = a_0(t) + \sum_{k=1}^{\infty}\sum_{j_1,\ldots,j_k\geq 1}\left[a_0\Big(t-\sum_{s=1}^{k} j_s\Big)\right]\prod_{r=1}^{k}\left[a_{j_r}\Big(t-\sum_{s=1}^{r-1} j_s\Big)\right]\prod_{r=1}^{k}\eta_{t-\sum_{s=1}^{r} j_s}^2$$

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$$\leq \left[\sup_n \frac{\omega_n}{\beta_n(1)}\right] + \left[\sup_n \frac{\omega_n}{\beta_n(1)}\right] \sum_{k=1}^{\infty} \sum_{j_1,\ldots,j_k \geq 1} \prod_{r=1}^{k} \left[\sup_n b_{n,j_r}\right] \prod_{r=1}^{k} \eta^2_{t-\sum_{s=1}^{r} j_s}$$

The inequality holds because that $\sum_{n=1}^{N}[I(s_t = n)] = 1$, then for all t we have that $\left[\sup_t a_0(t)\right] = \left[\sup_n \frac{\omega_n}{\beta_n(1)}\right]$ and $\left[\sup_t a_k(t)\right] = \left[\sup_n b_{n,k}\right]$ for all $k \geq 1$. As conditions (3) and (4) of the Assumption 1 of Dahlhaus et al. (2006) hold, the existence of the finite variance and the uniqueness of the above solution can be straightforwardly proved using the similar steps in their proof of the Proposition 1.

Additionally, given $p_{ij} \in (0,1)$, s_t is irreducible and aperiodic with unique stationary probabilities denoted by $\pi_n = P(s_t = n) = E(I(s_t = n))$ for all state n. Thus, $E(a_0(t)) = \sum_{n=1}^{N}(\pi_n\omega_n)/\beta_n(1)$.

Further, it can be shown that $E(I(s_{t-j} = r)I(s_t = s)) = p_{rs}^{(j)} = P(s_t = s|s_{t-j} = r)$, which does not depend on t. Using this result, we have that

$$E(a_0(t-j)a_0(t)) = E\left(\sum_{r=1}^{N} I(s_{t-j} = r)\frac{\omega_r}{1-\beta_{r,1}}\left[\sum_{s=1}^{N} I(s_t = s)\frac{\omega_s}{1-\beta_{s,1}}\right]\right)$$

Clearly, all products of the variables in the above equation has the form of $I(s_{t-j})I(s_t)$, the expectation of which only depends on j. Since for all $b, h \geq 0$, $E(a_g(t-j)a_h(t))$ is composed of weighted summations over such expectations, it will not depend on t.

As for more complex cases like $\prod_{r=1}^{k} I(s_{t-\sum_{s=1}^{r} j_s} = n_r)$, its expectation is only determined by the sample path of s_t during the period $(t-\sum_{s=1}^{k} j_s, t-j_1)$, which does not depend on t. More specifically, this expectation is equal to $p_{n_k,n_{k-1}}^{(j_k)} p_{n_{k-1},n_{k-2}}^{(j_{k-1})} \cdots p_{n_2,n_1}^{(j_2)}$. This is because of the Markov property, and only the lengths of time between transitions matter, rather than the exact time when transition occurs.

Therefore, it implies that the expectation of the second part of Equation (14) will also not depend on t, and hence $E(\varepsilon_t^2) = E(h_t)$ is a finite constant determined by ω_n, $\phi_{n,i}$, $\beta_{n,j}$, γ_n, d_n and all transition probabilities of s_t.

C. Proof of Theorem 2

As suggested by Li et al. (2015) that $E(\eta_t^4)^{-1/2} \leq E(\eta_t^2)^{-1} = 1$, all conditions of the Theorem 1 hold here and hence finite $E(\varepsilon_t^2)$ exists and does not depend on t.

By expanding h_t^2 with Equation (14) and using the fact that $I(s_t = n)^q = I(s_t = n)$ for any $q > 0$, it can be shown that all its variable components have similar forms to $\prod_{r=1}^{k} I(s_{t-\sum_{s=1}^{r} j_s} = n_r)$. Therefore, using the same argument made in Appendix B, $E(h_t^2)$ will not depend on t, neither will $E(\varepsilon_t^4) = E(\eta_t^4)E(h_t^2)$. Denote that $M_m = E(\varepsilon_t^{2m})$ and $\mu_m = E(\eta_t^{2m})$ for m =1 or 2, then $M_m = \mu_m E(h_t^m)$.

From Appendix B, we know that $h_t \leq h_t^*$ and $h_t^* = b_0^* + \sum_{k=1}^{\infty} b_k^* \varepsilon_{t-k}^2$ with $b_0^* = \left[\sup_n \frac{\omega_n}{\beta_n(1)}\right]$ and $b_k^* = \left[\sup_n b_{n,k}\right]$ for all $k \geq 1$. Thus, we have that

$$(h_t^*)^2 = (b_0^*)^2 + 2b_0^* \sum_{k=1}^{\infty} b_k^* \varepsilon_{t-k}^2 + \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} b_i^* b_j^* \varepsilon_{t-i}^2 \varepsilon_{t-j}^2$$

By the Hölder's inequality, it holds that $E(\varepsilon_{t-i}^2 \varepsilon_{t-j}^2) \leq M_2$. Hence,

$$M_2 = \mu_2 E(h_t^2) \leq \mu_2 \left\{(b_0^*)^2 + 2b_0^* \gamma^* M_1 + (\gamma^*)^2 M_2\right\}$$

where $\gamma^* = \sum_{k=1}^{\infty} b_k^*$. Further using the assumption that $\gamma^* < \mu_2^{-1/2}$, $(1 - \mu_2(\gamma^*)^2) > 0$

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and

$$M_2 \leq \frac{\mu_2}{1-\mu_2(\gamma^*)^2}\left\{(b_0^*)^2 + 2b_0^*\gamma^* M_1\right\}$$

Additionally, we have finite μ_2 and γ^* by assumption, $M_1 < \infty$ from Theorem 1 and $b_0^* < \infty$ as constraints of Conrad (2006) hold for all states, which prove $M_2 < \infty$.

Next, we consider $h_{t-j}h_t$ for all $j \geq 1$. By the Hölder's inequality, its expectation will be smaller than $E(h_t^2)$ and hence be finite. Using Equation (14), we have

$$\begin{aligned}
h_{t-j}h_t =& a_0(t-j)a_0(t) \\
&+ a_0(t)\sum_{k=1}^{\infty}\sum_{j_1,\ldots,j_k\geq 1}\left[a_0\Big(t-j-\sum_{s=1}^{k}j_s\Big)\right]\prod_{r=1}^{k}\left[a_{j_r}\Big(t-j-\sum_{s=1}^{r-1}j_s\Big)\right]\prod_{r=1}^{k}\eta^2_{t-j-\sum_{s=1}^{r}j_s} \\
&+ a_0(t-j)\sum_{k=1}^{\infty}\sum_{j_1,\ldots,j_k\geq 1}\left[a_0\Big(t-\sum_{s=1}^{k}j_s\Big)\right]\prod_{r=1}^{k}\left[a_{j_r}\Big(t-\sum_{s=1}^{r-1}j_s\Big)\right]\prod_{r=1}^{k}\eta^2_{t-\sum_{s=1}^{r}j_s} \\
&+ \left(\sum_{k=1}^{\infty}\sum_{j_1,\ldots,j_k\geq 1}\left[a_0\Big(t-j-\sum_{s=1}^{k}j_s\Big)\right]\prod_{r=1}^{k}\left[a_{j_r}\Big(t-j-\sum_{s=1}^{r-1}j_s\Big)\right]\prod_{r=1}^{k}\eta^2_{t-j-\sum_{s=1}^{r}j_s}\right) \\
&\left(\sum_{k=1}^{\infty}\sum_{j_1,\ldots,j_k\geq 1}\left[a_0\Big(t-\sum_{s=1}^{k}j_s\Big)\right]\prod_{r=1}^{k}\left[a_{j_r}\Big(t-\sum_{s=1}^{r-1}j_s\Big)\right]\prod_{r=1}^{k}\times\eta^2_{t-\sum_{s=1}^{r}j_s}\right)
\end{aligned}$$

When taking expectation, the first component only depends on j, as proved in Appendix B. In the second component, the essential part is

$$E\left(\left[\sum_{n=1}^{N}I(s_t=n)\right]\sum_{k=1}^{\infty}\sum_{j_1,\ldots,j_k\geq 1}\left[\sum_{n=1}^{N}I(s_{t-j-\sum_{s=1}^{k}j_s}=n)\right]\prod_{r=1}^{k}\left[\sum_{n=1}^{N}I(s_{t-j-\sum_{s=1}^{r-1}j_s}=n)\right]\right)$$

which only depends on the length of time spent between transitions. As $t > (t-j-j_s)$ for all s and $j \geq 1$, the last transition always occurs from $t-j-1$ to t, with a length of $j+1$. Therefore, since the limits of summations and products are all the same and are independent of t and j, such an expectation will be only determined by j.

The third component is more complex, with a similar essential part of expectation as shown above. The major difference is $(t-j) < t$, so that the time of transition from/to $t-j$ will depend on the difference between $t-j$ and $t-\sum_{s=1}^{k}j_s$. However, the resulting complete sample path of s_t still does not depend on t, and will be completely determined by the difference between j and $\sum_{s=1}^{k}j_s$. As j_s of Equation (14) is the same for all t, it will not depend on j. Consequently, this component will also only be determined by j.

With the finite μ_2 and constant parameters, the expectation of the last component will still be only affected by s_t. Despite its complexity, all the limits of summations and products are the same for all t and j. Thus, the sample path of s_t depends on the difference between j and j_s. Since the value of j_s is independent of j, this expectation will only depends on j.

Summing up the above arguments, finite $E(h_{t-j}h_t)$ exists and only depend on j. As $E(\varepsilon^2_{t-j}\varepsilon^2_t) = E(\eta^2_{t-j})E(\eta^2_t)E(h_{t-j}h_t) = E(h_{t-j}h_t)$ and finite M_1 exists, the covariance of ε_t^2 will be independent of t and vary only with j. Hence, ε_t^2 is covariance stationary, which completes the proof.

D. Proof of Lemma 1

The cases of $m = 1$ or 2 are implied by Theorems 1 and 2. For $m = 3$ or 4, expanding $E(h_t^m)$ with Equation (14) will have all variable components with similar

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form to $\prod_{r=1}^{k} I(s_{t-\sum_{s=1}^{r} j_s} = n_r)$. As demonstrated in Appendixes B and C, such forms are independent of t. This can be directly applied to $E(\varepsilon_t^{2m}) = E(\eta_t^{2m})E(h_t^m)$. Using the same notations employed in Appendix C, Hölder's inequality suggests that $E(\varepsilon_{t-i}^2\varepsilon_{t-j}^2\varepsilon_{t-k}^2) \leq M_3$ and $E(\varepsilon_{t-i}^2\varepsilon_{t-j}^2\varepsilon_{t-k}^2\varepsilon_{t-s}^2) \leq M_4$ for all i, j, k and $s \in \{1, 2, \ldots, \infty\}$.

With those results and following the same steps used in Appendix C, it is straightforward to show that

$$M_3 \leq \frac{\mu_3}{1-\mu_3(\gamma^*)^3}\left\{(b_0^*)^3 + 3(b_0^*)^2\gamma^* M_1 + 3b_0^*(\gamma^*)^2 M_2\right\}$$
$$M_4 \leq \frac{\mu_4}{1-\mu_4(\gamma^*)^4}\left\{(b_0^*)^4 + 4(b_0^*)^3\gamma^* M_1 + 6(b_0^*)^2(\gamma^*)^2 M_2 + 4b_0^*(\gamma^*)^3 M_3\right\}$$

Hence, with the assumptions $\gamma^* < \mu_3^{-1/3}$ ($\gamma^* < \mu_4^{-1/4}$), we can prove that $M_3 < \infty$ ($M_4 < \infty$).

For any finite m, using the same argument made above, it is straightforward to verify that the variable components of $E(h_t^m)$ is determined by s_t and thus does not depend on t. Moreover, $(h_t^*)^m$ can be expanded as binomial series. Further with Hölder's inequality, we can demonstrate that

$$M_m \leq \frac{\mu_m}{1-\mu_m(\gamma^*)^m}\left\{(b_0^*)^m + \binom{m}{1}(b_0^*)^{m-1}\gamma^* M_1 + \cdots + \binom{m}{1}b_0^*(\gamma^*)^{m-1}M_{m-1}\right\}$$

Applying the assumption $\gamma^* < \mu_m^{-1/m}$, we can complete the proof that $M_m < \infty$ for all finite $m \geq 1$.

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